

STEPHANOS LOIZOU

Passionate about working with data to create robust solutions to problems that surpass human capability. Experienced in delivering Computer Vision products in the robotics domain.

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PROFESSIONAL EXPERIENCE

- **Staff Engineer - Perception / Vision, Oxa, UK - April 2025 - Present**
 - Overseeing the deployment of multi-modal sensor-fusion ML models and components to edge compute devices.
 - Contributing to the design and implementation of end-to-end GPU-zero-copy robotics-oriented architecture.
- **Senior Engineer - Perception / Vision, Oxa, UK - November 2022 - April 2025**
 - Led Traffic Signal Mapping and Classification. Prioritising and delivering features, bug fixes and architectural improvements.
 - Improved online classification performance of Traffic Light Recognition system by more than 10% as well as increased coverage to new domains.
 - Deployed proof-of-concept multi-modal sensor-fusion ML models to edge devices (C++, Python, CUDA, TensorRT).
 - Implemented and automated metrics and report generation reducing iteration time by more than 5x.
 - R&D in drift detection methods and productisation scalable of MLOps pipelines.
 - Contributed to real-time cross-platform codebase designed to run on various platforms (C++)
 - Contributed to development of common ML tools and pipelines (Google Cloud, Python)
- **Senior Development Engineer (Full Time), Seebyte, UK - February 2020 - November 2022**
 - Led unsupervised segmentation product. Human-in-the-loop workflows, combining recent and classical ML approaches. ML product design, software architecture (Python, C++, Project Management, Client Facing)
 - R&D and Domain Specialisation and Adaptation of SOTA vision-based techniques. (Object Detection, Segmentation, Edge-devices, Confidence Estimation, Feature Extraction, Data compression, autoencoders). Good understanding of object detection architectures, CNN, VAE, and multitask learning. (Tensorflow, Pytorch, TensorRT, Python, C++)
 - Deliver AI-based (DL/ML) products to clients. Presenting technical findings to both technical and non-technical audiences.
 - Contributed to in-house MLOps framework to automate model training and validation reducing person-hours by more than 100% while improving QA. Contributed to Cross-Platform DevOps libraries (Jenkins, Groovy, CI, CT)
 - Worked on ETL pipelines adding features, curating data, robustification and validation.
 - Extended, Fixed and Unit-tested multiple components of internal tools, C++, Python. Regular Code and Design Reviews.
- **Development Engineer (Full Time) - Seebyte, UK June 2018-February 2020**
 - Implemented safety-critical components in underwater ROV control over 4G communications. State-machines, C++, low-latency drivers. (First 6 months)
 - Joined the Automated Target Recognition team. Implemented algorithms for object detection in SONAR Data. Improved key metrics by over 50% over previous approaches.
 - Contributed to ML workflow improvements and components of the in-house libraries and tools in data analysis and management.
- **Demonstrator, Tutor, Teaching Assistant, (Part-Time) - University of Edinburgh, UK - September 2016 - May 2018**
 - Weekly labs, tutorials and monitoring forums to respond to and assist students.
- **Data Analyst/Developer (Full-time Internship): - Annalect OMG, UK - June 2016- August 2016**
 - Research and implementation of methods in multi-touch attribution (advertising). Implemented and Compared packages in R. Presented to management.
- **Software Developer (Part-time / Freelance) - Dangerfarms, UK - Aug 2014 - August 2015**
 - Test-Driven Web Development. MVC, Django, AngularJS, Ionic HTML, CSS, and Javascript, RESTful APIs, Optimised for large databases, Unit tested production code.

EDUCATION / MILITARY TRAINING

- **Computer Science (MInf) - University of Edinburgh, Scotland - Sep 2013 to May 2018**
 - **Classification:** First Class with Honours in Master of Informatics

- **Directional Courses:** Robot Learning and Sensorimotor Control, Intelligent Autonomous Robotics, Machine Learning and Pattern Recognition, Advanced Vision and Robotics, Computational Cognitive Neuroscience, Applied Machine Learning, Algorithms and Data Structures, Computer Security, Linear Algebra, Calculus and Applications, Probability.
- **Second Lieutenant** - Cypriot National Guard - **Jun 2011 to Jul 2013**
 - Leader of a 22-soldier squad
 - Conscript and firearms speciality trainer

INTERESTING EARLY WORK

- Worked on a Quadruped-robot <https://youtu.be/VJd5niX49oU> (2018); **Adapted Proprioceptive State Estimate to platform.**
- Co-founded an Autonomous Formula Student Group <https://www.eufs.co/ai-team> (2017); **To today, the most awarded team in the UK**
- Built an Award-winning Football-playing ball-curving robot. <https://www.youtube.com/watch?v=bn2u76CltBg> (2016); **The one that made me fall in love with robotics.**
- My First “full-size” robot project <https://www.youtube.com/watch?v=osOMybg17qM> (2016); **Enabling rough terrain navigation for a wheeled platform. Traversability estimation.**

SOFTWARE / DEV SKILLS / FRAMEWORKS

- **Current:** Modern C++(std17+),Cmake, Boost, Python/PEP, Pytorch, TensorFlow/Keras,numpy, UNIX & Windows, Docker, Jenkins, OpenCV, Google Cloud
- **Past / Familiar:** ROS, PCL,Javascript, Java, Eigen, Django, Flask, HTML/CSS, URDF,Gazebo